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# Tumor necrosis factor-α reduces adiponectin production by decreasing transcriptional activity of peroxisome proliferator-activated receptor-γ in calf adipocytes

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## ABSTRACT

Adiponectin (encoded by *ADIPOQ*) is an adipokine that orchestrates energy homeostasis by modulating glucose and fatty acid metabolism in peripheral tissues. During the periparturient period, dairy cows often develop adipose tissue inflammation and decreased plasma adiponectin levels. Proinflammatory cytokine tumor necrosis factor-α (TNF-α) plays a pivotal role in regulating the endocrine functions of adipocytes, but whether it affects adiponectin production in calf adipocytes remains obscure. Thus, the present study aimed to determine whether TNF-α could affect adiponectin production in calf adipocytes and to identify the underlying mechanism. Adipocytes isolated from Holstein calves were differentiated and used for (1) BODIPY493/503 staining; (2) treatment with 0.1 ng/mL TNF-α for different times (0, 8, 16, 24, or 48 h); (3) transfection with peroxisome proliferator-activated receptor-γ (*PPARG*) small interfering RNA for 48 h followed by treatment with or without 0.1 ng/mL TNF-α for 24 h; and (4) overexpression of *PPARG* for 48 h followed by treatment with or without 0.1 ng/mL TNF-α for 24 h. After differentiation, obvious lipid droplets and secretion of adiponectin were observed in adipocytes. Treatment with TNF-α did not alter mRNA abundance of *ADIPOQ* but reduced the total and high molecular weight (HMW) adiponectin content in the supernatant of adipocytes. Quantification of mRNA abundance of endoplasmic reticulum (ER)/Golgi resident chaperones involved in adiponectin assembly revealed that ER protein 44 (*ERP44*), ER oxidoreductase 1α (*ERO1A*), and disulfide bond-forming oxidoreductase A-like protein (*GSTK1*) were downregulated in TNF-α-treated adipocytes, while 78-kDa glucose-regulated protein and Golgi-localizing γ-adaptin ear homology domain ARF binding protein-1 were unaltered. Moreover, TNF-α diminished nuclear translocation of PPARγ and downregulated mRNA abundance of *PPARG* and its downstream target gene fatty acid synthase, suggesting that TNF-α suppressed the transcriptional activity of PPARγ. In the absence of TNF-α, overexpression of *PPARG* enhanced the total and HMW adiponectin content in supernatant and upregulated the mRNA abundance of *ADIPOQ*, *ERP44*, *ERO1A*, and *GSTK1* in adipocytes. However, knockdown of *PPARG* reduced the total and HMW adiponectin content in supernatant and downregulated the mRNA abundance of *ADIPOQ*, *ERP44*, *ERO1A*, and *GSTK1* in adipocytes. In the presence of TNF-α, overexpression of *PPARG* decreased, while knockdown of *PPARG* further exacerbated TNF-α-induced reductions in total and HMW adiponectin secretion and gene expression of *ERP44*, *ERO1A*, and *GSTK1*. Overall, TNF-α reduces adiponectin assembly in the calf adipocyte, which may be partly mediated by attenuation of PPARγ transcriptional activity. Thus, locally elevated levels of TNF-α in adipose tissue may be one reason for the decrease in circulating adiponectin in periparturient dairy cows.

**Key words:** TNF-α, adiponectin, PPARγ, transition period

## INTRODUCTION

Adiponectin (encoded by *ADIPOQ*) is almost exclusively secreted by adipocytes and is one of the most abundant adipokines in the serum of adult dairy cows (Singh et al., 2014a; Mann et al., 2018). Previous studies in mice have reported that adiponectin improves systematic insulin sensitivity and alleviates hepatic lipid accumulation (Lee and Lee, 2015; Kuramoto et al., 2021). In dairy cows, serum concentration of adiponectin was positively associated with a revised quan-

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titative insulin sensitivity check index during the early lactation period (Singh et al., 2014b). In addition, adiponectin promoted lipid oxidation and suppressed lipid synthesis in bovine hepatocytes (Chen et al., 2013). Of note, blood adiponectin falls to a nadir at calving and recovers during early lactation in dairy cows (Giesy et al., 2012). However, the mechanisms regulating transcription, assembly, and secretion of adiponectin in calf adipocytes are still poorly understood.

At least in nonruminants, adiponectin is synthesized as a 30-kDa monomer, which is then posttranslationally modified to assemble into multiple multimeric forms, including low molecular weight (**LMW**), middle molecular weight (**MMW**), and high molecular weight (**HMW**) forms, before secretion (Waki et al., 2003; Fang and Judd, 2018). Of note, multiple studies have indicated that the HMW adiponectin appears to be the most biologically active (Pajvani et al., 2004; Wang et al., 2008). Assembly of adiponectin occurs in the endoplasmic reticulum (**ER**) and Golgi and depends on the interaction of ER/Golgi resident chaperones 78-kDa glucose-regulated protein (**GRP78**; encoded by *HSPA5*), ER protein 44 (**ERP44**), ER oxidoreductase 1α (**Ero1-Lα**; encoded by *ERO1A*), disulfide bond-forming oxidoreductase A-like protein (**DsbA-L**; encoded by *GSTK1*), and Golgi-localizing γ-adaptin ear homology domain ARF binding protein-1 (**GGA1**; Xie et al., 2006; Wang and Scherer, 2008). Although downregulated expression of *HSPA5* and *ERP44* in adipose tissue (**AT**) and decreased content of serum HMW adiponectin were observed in dairy cows during early lactation, the mRNA abundance of *ADIPOQ* was not altered in AT (Lemor et al., 2009; Giesy et al., 2012). Thus, it is reasonable to speculate that impaired adiponectin assembly in adipocytes may be responsible for reduced adiponectin content in the blood of dairy cows after parturition.

Peroxisome proliferator-activated receptor-γ (**PPARγ**), a transcription factor highly expressed in adipocytes, is a master regulator of adipogenesis and lipid storage (Christodoulides and Vidal-Puig, 2010; Cai et al., 2016). In mice and humans, adiponectin transcription is tightly regulated by PPARγ (Maeda et al., 2001; Iwaki et al., 2003). Further, given that overexpression of *PPARG* upregulated the expression of *ERO1A* and *GSTK1* in differentiated mouse embryo fibroblast (3T3-L1) adipocytes (He et al., 2016), PPARγ may also be involved in adiponectin assembly. Interestingly, Schmitt et al. (2011) found that the mRNA expression of *PPARG* in AT of postpartum cows was downregulated, which underscored a potential link between decreased PPARγ transcriptional activity and reduced blood adiponectin levels.

A growing body of evidence indicates that AT inflammation often develops in postpartum dairy cows (Vailati-Riboni et al., 2017; Newman et al., 2019). Adipocytes and AT macrophages are the major producers of tumor necrosis factor-α (TNF-α) under inflammatory conditions (Mukesh et al., 2010; Olefsky and Glass, 2010), and TNF-α was reported to reduce adiponectin secretion in human and murine adipocytes (Simons et al., 2007; He et al., 2016). Moreover, TNF-α decreased gene expression and nuclear translocation of PPARγ in C2C12 mouse myoblasts (Magee et al., 2012). Considering the increased serum level of TNF-α and decreased abundance of *PPARG* in AT of postpartum dairy cows (Schmitt et al., 2011; Mecitoglu et al., 2016), we hypothesized that TNF-α reduces adiponectin production by regulating the transcriptional activity of PPARγ in calf adipocytes. We thus investigated the effects of TNF-α on PPARγ transcriptional activity and adiponectin transcription, assembly, and secretion in calf adipocytes. Furthermore, we also assessed the role of PPARγ in regulating adiponectin production.

## MATERIALS AND METHODS

### Ethics

All animal procedures were approved by the Ethics Committee on the Use and Care of Animals of Jilin University (Changchun, China, SY202208004).

### Sample Size Determination

Based on the changes in circulating adiponectin concentrations in dairy cows during early lactation (Giesy et al., 2012), we aimed to detect a 0.45-fold difference in adiponectin content in the supernatant of adipocytes between control and TNF-α-treated groups. To ensure that the study had a 95% power to detect this difference as statistically significant ($P = 0.01$), 5 calves per group were required as calculated by the Boston University Research Support online tool IACUC (https://www.bu.edu/researchsupport/compliance/animal-care/working-with-animals/research/sample-size-calculations-iacuc/, 2018 revision).

### Isolation of Primary Preadipocytes

Isolation of primary preadipocytes from 5 healthy female Holstein calves (1 d old) with similar BW (median = 37.6 kg, range = 34.8–40.3 kg) was performed according to previously published methods with minor modifications (Yin et al., 2015). Adipose tissue from the omentum was obtained surgically under sterile con-

ditions and rinsed with sterile PBS containing penicillin (2,500 U/mL) and streptomycin (2,500 μg/mL) to remove adherent blood. The fascia and visible blood vessels in the tissue were peeled away, and AT was cut into small pieces of approximately 1 $mm^3$. The resulting AT was digested with Dulbecco's modified Eagle's medium (**DMEM**)/F12 (SH30023.01; HyClone) digestion solution containing 1 mg/mL collagenase type I (C0130; Sigma-Aldrich Co.) at 37°C and incubated in a shaking water bath for 1.5 h. Then, the mixture was filtered through 80- and 40-μm cell filters in sequence. The collected filtrate was separated from adipocytes and medium by centrifugation at 175 × *g* for 10 min at room temperature. The residual erythrocytes in the resulting cell pellet were removed by adding ammonium-chloride-potassium lysis buffer (C3702; Beyotime Institute of Biotechnology) for 2 min at room temperature, followed by centrifugation at 175 × *g* for 10 min at room temperature. Subsequently, the supernatant was discarded, and remaining cells were resuspended in basal culture medium (**BCM**) composed of DMEM/F12 with 10% fetal bovine serum (SH30084.03; HyClone) and 1% penicillin-streptomycin (Sv30010; HyClone). After cell counting, the cell suspension was adjusted to a concentration of $1 \times 10^4$ cells/mL and inoculated in a cell culture flask. Preadipocytes were then incubated at 37°C in a humidified atmosphere with 5% $CO_2$ in an incubator with BCM for 24 h, after which the medium was replaced to remove nonadherent cells and tissue residues. Then, BCM was replaced every other day until the next experiment.

### Preadipocyte Differentiation

Preadipocytes were inoculated in 6-well cell culture plates (Corning Costar Corp.) and cultured in BCM. After cells were approximately 70% confluent, the BCM was replaced with freshly prepared differentiation culture medium (**DCM**)1 to induce differentiation (0 d). The DCM1 was prepared by adding 3-isobutyl-1-methylxanthin (I-7018, Sigma-Aldrich), dexamethasone (D-4902, Sigma-Aldrich), and insulin (I-5500, Sigma-Aldrich) to the BCM at final concentrations of 0.5 m*M*, 1 μ*M*, and 1 μg/mL, respectively. After 48 h, DCM1 was replaced with DCM2 to maintain the differentiation culture. The DCM2 was prepared by adding insulin at a final concentration of 1 μg/mL to the BCM. Fresh DCM2 was replaced every 48 h for 10 d until visible lipid droplets appeared in the cell, a sign that adipocytes were differentiated. Cell quantification was determined by manually using a hemocytometer. After 12 d of differentiation, the total amount of adipocytes was $3.90 \times 10^5$ to $4.15 \times 10^5$ per well of a 6-well plate with an average of $4 \times 10^5$ cells.

### Cell Treatment

A schematic diagram of the timeline of the cell culture and treatment is shown in Supplemental Figure S1 (https://doi.org/10.6084/m9.figshare.21992594.v11; Du et al., 2023). Specifically, experiments were divided into 4 sections. First, BODIPY493/503 staining was performed at 0 and 12 d after differentiation, and supernatant and cells were collected at 0, 4, 8, and 12 d after differentiation. Second, adipocytes were treated with recombinant bovine TNF-α (0.1 ng/mL; ab87904, Abcam) for 0, 8, 16, 24, and 48 h after 12 d of differentiation, and supernatant and cells were then collected. The choice of TNF-α dose was based on blood concentration of TNF-α in dairy cows during the early postpartum period (Mecitoglu et al., 2016; El-Deeb and El-Bahr, 2017). Third, to silence *PPARG*, adipocytes were transfected with 100 pmol of *PPARG* small interfering (**si**) RNA (**siPPARG**; sense 5′-GCCCAUUGAGGACAUACAATT-3′ and antisense 5′-UUGUAUGUCCUCAAUGGGCTT-3′) or scrambled negative control siRNA (siControl; sense 5′-UUCUCCGAACGUGUCACGUTT-3′ and antisense 5′-ACGUGACACGUUCGGAGAATT-3′) with Lipofectamine 2000 (11668019, Invitrogen) for 48 h after 12 d of differentiation. Following treatment with or without 0.1 ng/mL TNF-α for 24 h, supernatant and cells were collected. In this study, transfection with siPPARG achieved 80.9% gene silencing efficiency at the mRNA level. The specific siRNA for bovine *PPARG* was designed based on the bovine *PPARG* mRNA sequence (NM_181024.2) and was synthesized by GenePharma. Fourth, to overexpress *PPARG*, adipocytes were transfected with 4 μg of pShuttle-EGFP-PPARγ, a *PPARG* overexpression vector, or pShuttle-EGFP (p43312; Miaoling Bioscience), an empty vector used as a negative control, with Lipofectamine 2000 for 48 h after 12 d of differentiation. Following treatment with or without 0.1 ng/mL TNF-α for 24 h, supernatant and cells were collected. The plasmid pShuttle-EGFP was purchased from Miaoling Bioscience. Full-length *PPARG* (NM_181024.2) was amplified by PCR and subcloned into the pShuttle-EGFP vector to construct pShuttle-EGFP-PPARγ as previously reported (Jiang et al., 2020). All experiments were repeated in 5 calves and at least 3 technical replicates were performed in each calf.

### Sample Harvesting

For the fluorescent samples, cells were washed with PBS and fixed with 4% paraformaldehyde at 0 and 12 d of differentiation. After staining with BODIPY493/503 (D3922; Thermo Fisher Scientific) and 4′,6-diamidino-

2-phenylindole (**DAPI**; D9542, Sigma-Aldrich), the coverslips were sealed with glycerol. For the supernatant samples, after centrifugation at 350 × *g* for 10 min at room temperature, the collected supernatant was frozen in liquid nitrogen and stored at −80°C until analysis. For the cell samples used for gene abundance detection, cells were washed with PBS and lysed with RNAiso Plus (9109; TaKaRa Biotechnology Co. Ltd.) for RNA extraction. For the cell samples used for protein abundance detection, cells were centrifuged 1,000 × *g* for 5 min at 4°C and the cell pellets were used for subsequent protein extraction.

### Lipid Droplet Staining

Determining intracellular lipid accumulation was based on using BODIPY493/503 dye for fluorescent probes (Kaushik and Cuervo, 2015). Preadipocytes were inoculated in laser confocal dishes (Cat. No. 801002, NEST). At 0 and 12 d after differentiation, cells were washed with PBS and fixed with 4% paraformaldehyde for 20 min. Subsequently, cells were stained with BODIPY493/503 (1 μg/mL) for 15 min at room temperature followed by washing with PBS. Then, nuclei were stained with DAPI for 10 min. Coverslips were sealed with glycerol and samples were imaged using laser confocal microscopy (FV1200, Olympus). Pre-experiments were performed to avoid saturation of pixels, including selection of appropriate BODIPY493/503 dye treatment concentration and time and optimization of laser parameters.

### Quantitative Real-Time PCR Analysis

Total RNA was isolated from adipocytes using RNAiso Plus according to the manufacturer's instructions. The RNA concentration and quality were measured using a Nanophotometer N50 Touch (Implen GmbH) and by electrophoresis (1% agarose gels). Only the optical density 260:280 ratio of RNA in the range of 1.8 to 2.0 meets the requirements of MIQE guidelines for RNA purity (Bustin et al., 2009). In this study, the optical density 260:280 ratio of RNA samples ranged from 1.85 to 1.98. The concentration of purified RNA was evaluated by UV spectrum at 260 nm. Then, total RNA was reverse-transcribed into cDNA using a reverse transcription kit (RR047A, TaKaRa Biotechnology Co. Ltd.) according to the manufacturer's instructions. Quantitative real-time PCR was performed on a 7500 Real-Time PCR System (Applied Biosystems Inc.) using the SYBR Green plus reagent kit (RR420A, TaKaRa Biotechnology Co. Ltd.) to analyze the relative mRNA abundance of target genes. The amount of cDNA per reaction was 100 ng. The reaction conditions were as follows: 95°C for 30 s, followed by 45 cycles of 95°C for 5 s and 60°C for 34 s. Each sample was run in triplicate. The relative transcription of each target gene was normalized against the geometric mean of β-actin and 18S ribosomal RNA that were stably expressed in different groups (Supplemental Figure S2, https://doi.org/10.6084/m9.figshare.21992594.v11; Du et al., 2023). Gene expression was calculated with the $2^{-\Delta\Delta CT}$ method. The primer pairs used in this study were designed using Primer Express software 3.0 (Applied Biosystems Inc.) according to gene sequences published in GenBank (https://www.ncbi.nlm.nih.gov/genbank/), and they are listed in Supplemental Table S1 (https://doi.org/10.6084/m9.figshare.21992594.v11; Du et al., 2023). The specificity of primers was verified by Basic Local Alignment Search Tool searching at National Center for Biotechnology Information (https://www.ncbi.nlm.nih.gov/), and the quality of the primers was evaluated by agarose gel electrophoresis (a single band of correct size) and melt curve (a single peak). The PCR efficiency (E, 95% < E < 105%) and the correlation coefficient ($R^2$, higher than 0.99) were also used to assess the primers (Supplemental Table S1; Kubista et al., 2006).

### Protein Extraction and Western Blot Analysis

Total protein from adipocytes was extracted using a commercial protein extraction kit containing lysate buffers, phosphatase inhibitors, and protease inhibitors (C510003; Sangon Biotech Co. Ltd.) according to the manufacturer's instructions. Nuclear and cytoplasmic protein were extracted from adipocytes using a commercial kit containing nuclear and cytoplasmic extraction buffers (P1200; Applygen Technologies Inc.) according to the manufacturer's instructions. Protein concentrations were quantified using the bicinchoninic acid Protein Assay Kit (P1511, Applygen Technologies Inc.). Twenty micrograms of protein from each sample was separated by 12 or 15% SDS-PAGE and electrophoretically transferred onto 0.45-μm polyvinylidene fluoride membranes blocked in Tris-buffered saline solution with 0.01% Tween-20 (**TBS-T**) containing 3% BSA for 4 h at room temperature. Subsequently, blocked membranes were incubated overnight at 4°C with primary antibodies against PPARγ (bs-4590R, Bioss Biotechnology Co. Ltd.; 1:1,000), histone H3 (4499, Cell Signaling Technology Inc.; 1:1,000), β-actin (ab8226, Abcam; 1:2,000), β-tubulin (10094-1-AP, Proteintech; 1:1,000). After being washed 3 times with TBS-T, the membranes were incubated with horseradish peroxidase-conjugated anti-mouse

or anti-rabbit secondary antibody (Boster Biological Technology Co. Ltd.) for 45 min at room temperature. Immunoreactive bands were visualized using an enhanced chemiluminescence reagent (WBKLS0500, Millipore) via a protein imager (ProteinSimple). Last, all bands were quantified by calculating the integrated optical density from the area and optical density of each protein band using Image-Pro Plus 6.0 (Media Cybernetics Inc.) according to the manufacturer's instructions. Background subtraction was performed during quantification. Total protein abundance of PPARγ was standardized by β-actin, nuclear protein abundance of PPARγ was standardized by histone H3, and cytoplasmic protein abundance of PPARγ was standardized by β-tubulin. For the analysis of adiponectin multimers in the supernatant of adipocytes, 10-μL supernatant samples collected after centrifugation at 350 × *g* for 10 min at room temperature were subjected to 2 to 15% SDS-PAGE under nonreducing and non-heat-denaturing conditions as previously described (Waki et al., 2003). Subsequently, Western blot analysis was performed as described above by using anti-C-terminal globular domain antibody against adiponectin (ab181699, Abcam; 1:1,000). Protein abundance of HMW adiponectin in supernatant was standardized by intracellular β-actin.

### ELISA

The concentrations of total adiponectin in the supernatant of adipocytes were measured using a bovine adiponectin ELISA kit (SEKB-0196; Solarbio Science and Technology Co. Ltd.) according to the manufacturer's instructions. The detection range was 62 to 4,000 pg/mL. The supernatant was diluted 50-fold to reach the detection range. The intra- and inter-coefficients of variability of adiponectin were 7.8 and 10.5%, respectively. Sample absorbance values were read at 450 nm using a spectrophotometer (51119100, Thermo Fisher Scientific).

### Statistical Analysis

All data were analyzed using GraphPad Prism 8.0 (GraphPad Software Inc.) or SPSS 23.0 software (IBM Corp.). All data were tested for normality and homoscedasticity using the Shapiro-Wilk and Levene tests, respectively. One-way ANOVA or 2-way ANOVA with subsequent Bonferroni correction were performed for multiple comparisons. Linear and quadratic contrasts were conducted to evaluate time-dependent effects. The results are expressed as the mean ± standard error of the mean, and $P < 0.05$ was considered statistically significant.

## RESULTS

### Capacity of Calf Adipocytes to Secrete Adiponectin

Compared with results from 0 d of differentiation, BODIPY493/503 staining showed obvious accumulation of lipid droplets in calf adipocytes after 12 d of differentiation (Figure 1A). The mRNA abundance of the adipogenic markers CCAAT/enhancer-binding protein-α (*CEBPA*) and *PPARG* were upregulated in a linear and quadratic way with the increase of differentiation time and peaked at d 8 and 4 of differentiation, respectively (Figure 1B, C, $P < 0.05$; Supplemental Table S2, https://doi.org/10.6084/m9.figshare.21992594.v11; Du et al., 2023). The mRNA and supernatant content of adiponectin were undetectable at 0 d of differentiation but increased in a linear and quadratic fashion with increasing differentiation time (Figure 1D, E, $P < 0.05$; Supplemental Table S2).

### Effects of TNF-α on Adiponectin Transcription, Assembly, and Secretion

Compared with the control group, TNF-α (0.1 ng/mL) treatment had no effect on *ADIPOQ* mRNA abundance in adipocytes (Figure 2A, $P > 0.05$). The total and HMW adiponectin content in the supernatant of adipocytes was linearly reduced as a function of incubation time with TNF-α, with a nadir in response at 24 h (Figure 2B–D, $P < 0.05$; Supplemental Table S3, https://doi.org/10.6084/m9.figshare.21992594.v11; Du et al., 2023). The nuclear and cytoplasmic protein abundance of PPARγ in TNF-α-treated adipocytes showed a linear decrease with increasing incubation time (Figure 2E–G, $P < 0.05$; Supplemental Table S3). Furthermore, we observed a quadratic effect on the nuclear protein abundance of PPARγ in response to incubation time with TNF-α (Supplemental Table S3). In addition, a linear decrease for mRNA abundance of *PPARG* and its downstream target gene fatty acid synthase (*FASN*) occurred in TNF-α-treated adipocytes with the increase of incubation time (Figure 2H, I, $P < 0.05$; Supplemental Table S3). Treatment with TNF-α linearly downregulated mRNA abundance of adiponectin assembly regulators, *ERP44*, *ERO1A*, and *GSTK1* (Figure 2J-L, $P < 0.05$; Supplemental Table S3), but did not affect the mRNA abundance of *HSPA5* and *GGA1* (Figure 2M, N, $P > 0.05$) in adipocytes.

### Effects of Knockdown of PPARG on Adiponectin Production in TNF-α-Treated Adipocytes

Compared with the siControl group, siPPARG downregulated the mRNA abundance of *PPARG* and *FASN*

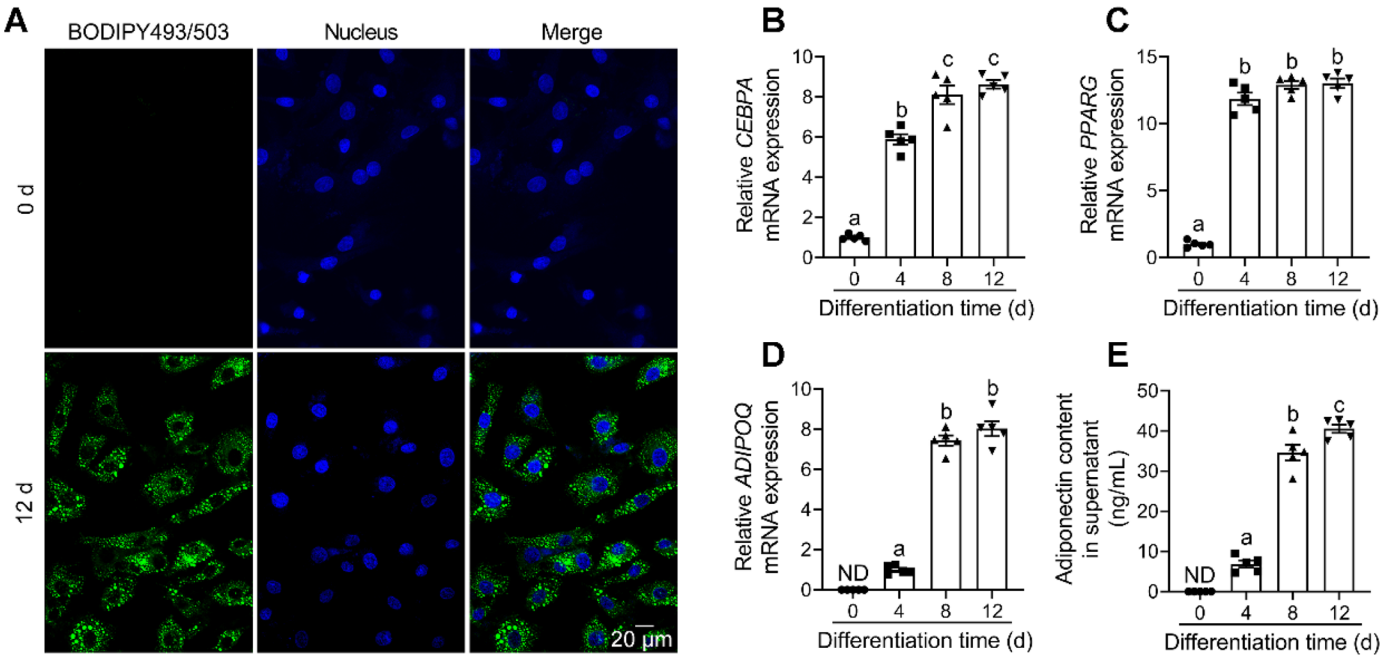



**Figure 1.** Capacity of calf adipocytes to secrete adiponectin. Pre-adipocytes were cultured in differentiation culture medium for 0 to 12 d. (A) Lipid drops were stained with BODIPY493/503 (green) and nuclei were stained with 4′,6-diamidino-2-phenylindole (blue) in calf adipocytes at 0 and 12 d after differentiation. Objective = ×20. Scale bar = 20 μm. (B, C, D) Relative mRNA abundance of CCAAT/enhancer-binding protein-α (*CEBPA*) and peroxisome proliferator-activated receptor-γ (*PPARG*) in calf adipocytes and expressed as fold change relative to the control group (0 d of differentiation). Relative mRNA abundance of adiponectin (*ADIPOQ*) in calf adipocytes and expressed as fold change relative to the control group (4 d of differentiation). ND = not detectable. (E) The content of adiponectin in the supernatant of adipocytes. Data are expressed as the means ± SEM. Data were analyzed by one-way ANOVA with subsequent Bonferroni correction. The same letter (a–c) indicates no significant difference ($P > 0.05$), whereas different letters indicate a significant difference ($P < 0.05$).

(Figure 3A, B, $P < 0.05$) and decreased total, nuclear, and cytoplasmic protein abundance of PPARγ (Figure 3C–F, $P < 0.05$). The mRNA abundance of *ADIPOQ* was downregulated in adipocytes transfected with siPPARG (Figure 3G, $P < 0.05$). Knockdown of *PPARG* diminished the total and HMW adiponectin content in the supernatant of adipocytes (Figure 3H–J, $P < 0.05$). In addition, knockdown of *PPARG* downregulated mRNA abundance of *ERP44*, *ERO1A*, and *GSTK1* (Figure 3K–M, $P < 0.05$).

In the presence of TNF-α, knockdown of *PPARG* further downregulated mRNA abundance of *PPARG* and *FASN* (Figure 3A, B, $P < 0.05$) and decreased total, nuclear, and cytoplasmic protein abundance of PPARγ (Figure 3C–F, $P < 0.05$). In cells transfected with siPPARG, TNF-α did not alter the mRNA abundance of *ADIPOQ* (Figure 3G, $P > 0.05$). Knockdown of *PPARG* exacerbated the reduction of total and HMW adiponectin content in the supernatant of TNF-α-treated adipocytes (Figure 3H–J, $P < 0.05$). Furthermore, in the presence of TNF-α, knockdown of *PPARG* also aggravated the downregulation of mRNA abundance of *ERP44*, *ERO1A*, and *GSTK1* induced by TNF-α (Figure 3K–M, $P < 0.05$).

### *Effects of Overexpression of PPARG on Adiponectin Production in TNF-α-Treated Adipocytes*

Compared with the pShuttle-EGFP group, overexpression of pShuttle-EFGP-PPARγ upregulated the mRNA abundance of *PPARG* and *FASN* (Figure 4A, B, $P < 0.05$) and increased total, nuclear, and cytoplasmic protein abundance of PPARγ (Figure 4C–F, $P < 0.05$). The mRNA abundance of *ADIPOQ* was upregulated in adipocytes overexpressing *PPARG* (Figure 3G, $P < 0.05$). Overexpression of *PPARG* increased the total and HMW adiponectin content in the supernatant of adipocytes (Figure 4H–J, $P < 0.05$). Additionally, mRNA abundance of *ERP44*, *ERO1A*, and *GSTK1* was upregulated in adipocytes overexpressing *PPARG* (Figure 4K–M, $P < 0.05$).

In the presence of TNF-α, overexpression of *PPARG* ameliorated TNF-α-induced downregulation in mRNA abundance of *PPARG*, *FASN*, *ERP44*, *ERO1A*, and *GSTK1* (Figure 4A, B, and K–M, $P < 0.05$), as well as the reduction in total, nuclear, and cytoplasmic protein abundance of PPARγ (Figure 4C–F, $P < 0.05$) and the total and HMW adiponectin content in the supernatant (Figure 4H–J, $P < 0.05$). In cells overexpressed

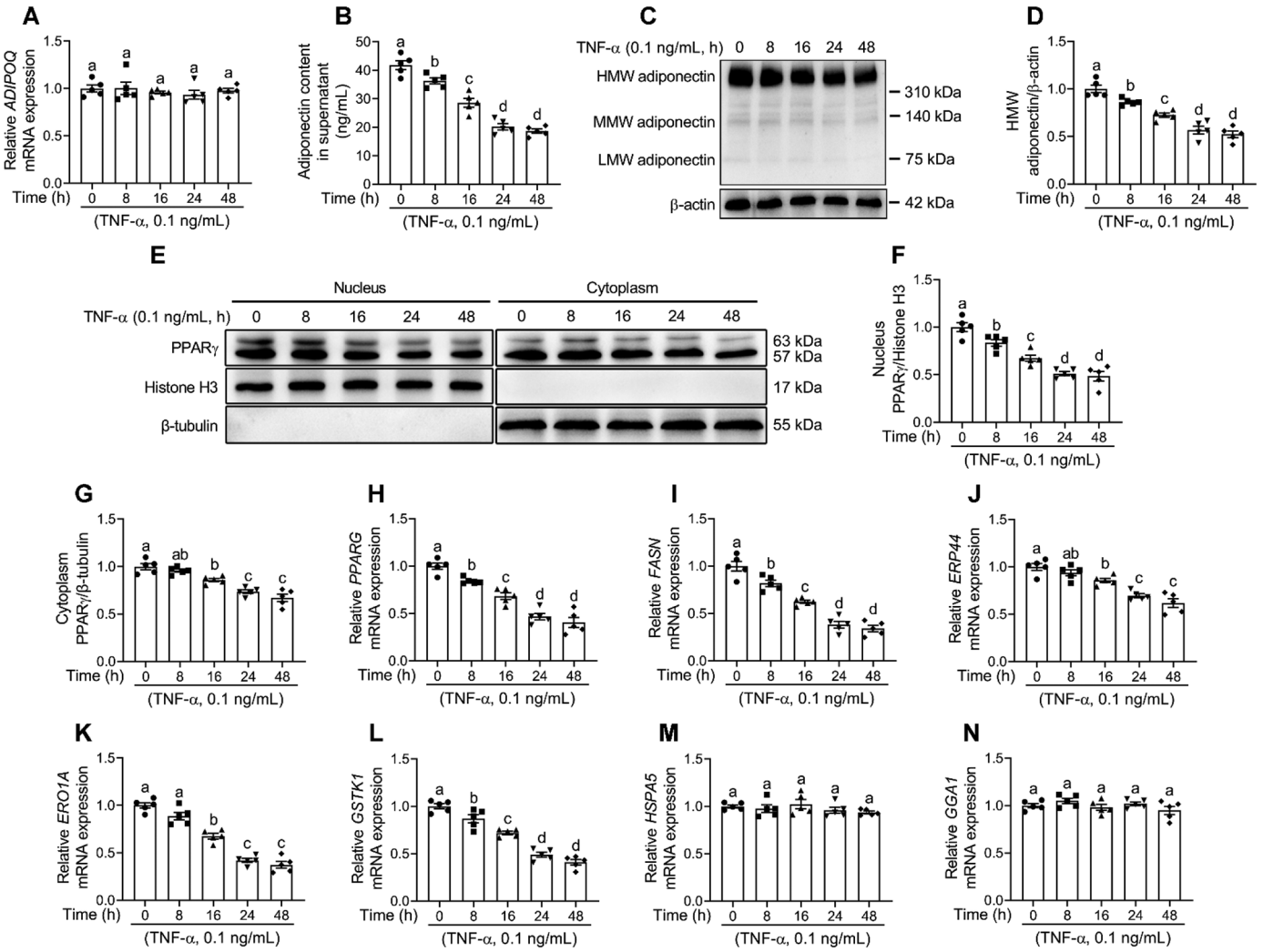



**Figure 2.** Effects of tumor necrosis factor-α (TNF-α) on adiponectin production. Adipocytes were treated with 0.1 ng/mL TNF-α for 0, 8, 16, 24, and 48 h after 12 d of differentiation. (A) Relative mRNA abundance of adiponectin (*ADIPOQ*) in calf adipocytes and expressed as fold change relative to the control group (no treatment). (B) The content of adiponectin in the supernatant of adipocytes. (C) Molecular weight distribution of adiponectin in the supernatant of adipocytes. HMW = high molecular weight; MMW = middle molecular weight; LMW = low molecular weight. (D) Quantification of the of HMW adiponectin protein level in the supernatant of adipocytes. (E) Representative blots of nuclear and cytoplasmic peroxisome proliferator-activated receptor-γ (PPARγ). (F, G) Quantification of the nuclear and cytoplasmic protein levels of PPARγ. (H–N) Relative mRNA abundance of *PPARG*, fatty acid synthase (*FASN*), endoplasmic reticulum protein 44 (*ERP44*), ER oxidoreductase 1α (*ERO1A*), disulfide bond-forming oxidoreductase A-like protein (*GSTK1*), 78-kDa glucose-regulated protein (*HSPA5*), and Golgi-localizing γ-adaptin ear homology domain ARF binding protein-1 (*GGA1*) in calf adipocytes and expressed as fold change relative to the control group (no treatment). Data are expressed as the means ± SEM. Data were analyzed by one-way ANOVA with subsequent Bonferroni correction. The same letter (a–e) indicates no significant difference ($P > 0.05$), whereas different letters indicate a significant difference ($P < 0.05$).

*PPARG*, TNF-α did not alter of the mRNA abundance of *ADIPOQ* (Figure 4G, $P > 0.05$).

## DISCUSSION

Inflammation of AT and reduced circulating adiponectin levels are often observed in dairy cows during early lactation (Singh et al., 2014b; Contreras et al., 2018). Although the role of adiponectin in different cells of dairy cows has been extensively studied (Chen et al., 2013; Kabara et al., 2014), little is known about the mechanisms that regulate adiponectin production. In this study, TNF-α decreased PPARγ transcriptional activity and adiponectin production in calf adipocytes, whereas overexpression of *PPARG* attenuated the inhibitory effects of TNF-α on adiponectin production (Figure 5). Therefore, decreased PPARγ transcriptional activity as a result of TNF-α challenge may potentially

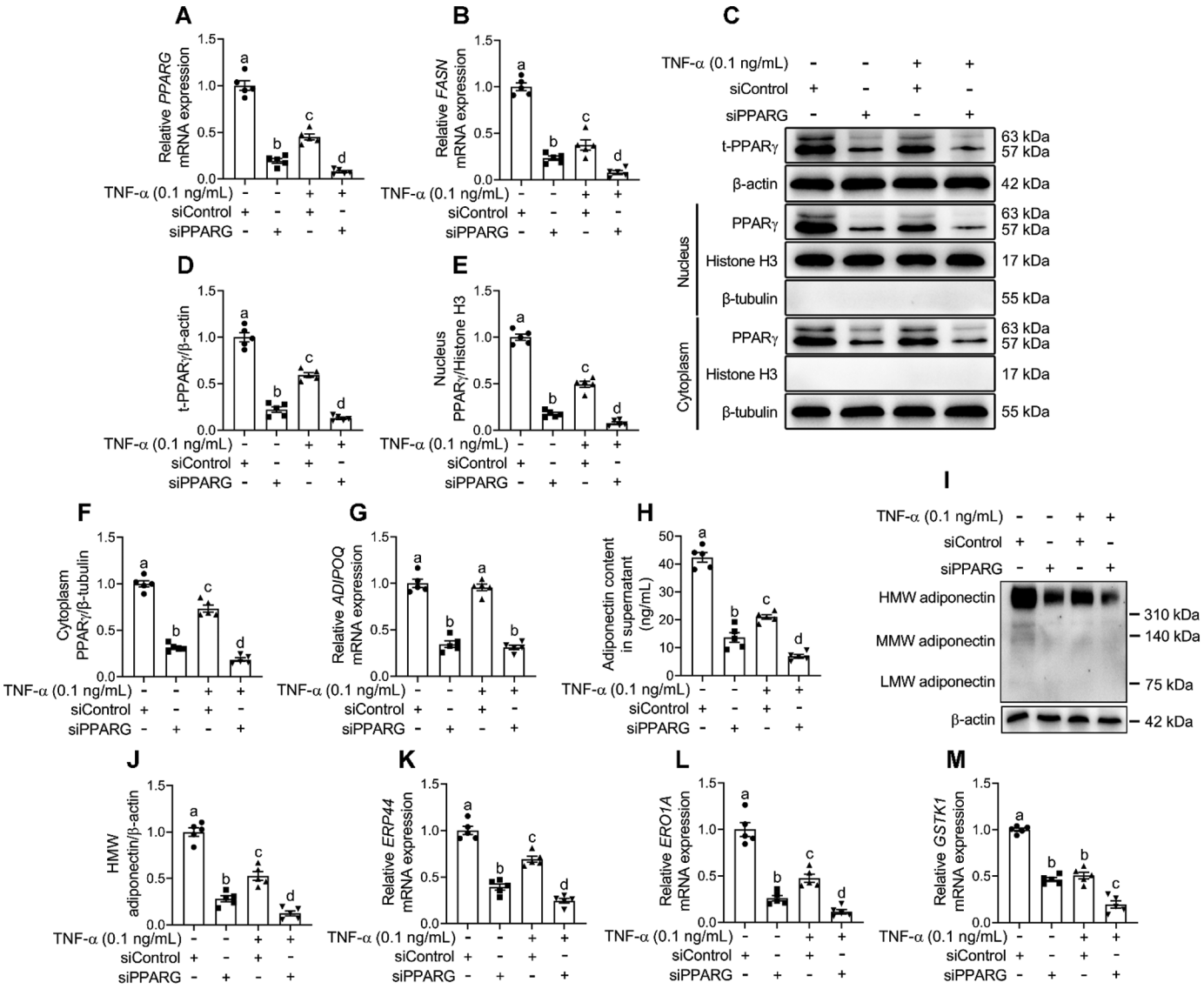



**Figure 3.** Effects of knockdown of peroxisome proliferator-activated receptor-γ (*PPARG*) on adiponectin production in tumor necrosis factor-α (TNF-α)-treated calf adipocytes. Adipocytes were transfected with *PPARG* small interfering (si) RNA (siPPARG) or negative control siRNA (siControl) for 48 h after 12 d of differentiation, and then treated with or without 0.1 ng/mL TNF-α for 24 h. (A, B) Relative mRNA abundance of *PPARG* and fatty acid synthase (*FASN*) in calf adipocytes and expressed as fold change relative to the control group (transfection with siControl). (C) Representative blots of total (t), nuclear, and cytoplasmic PPARγ. (D–F) Quantification of protein levels of t-PPARγ, nuclear PPARγ, and cytoplasmic PPARγ. (G) Relative mRNA abundance of adiponectin (*ADIPOQ*) in calf adipocytes and expressed as fold change relative to the control group (transfection with siControl). (H) The content of adiponectin in the supernatant of adipocytes. (I) Molecular weight distribution of adiponectin in the supernatant of adipocytes. HMW = high molecular weight; MMW = middle molecular weight; LMW = low molecular weight. (J) Quantification of the of HMW adiponectin protein level in the supernatant of adipocytes. (K–M) Relative mRNA abundance of endoplasmic reticulum protein 44 (*ERP44*), ER oxidoreductase 1α (*ERO1A*), and disulfide bond-forming oxidoreductase A-like protein (*GSTK1*) in calf adipocytes and expressed as fold change relative to control (transfection with siControl) group. Data are expressed as the means ± SEM. Data were analyzed by 2-way ANOVA with subsequent Bonferroni correction. The same letter (a–d) indicates no significant difference ($P > 0.05$), whereas different letters indicate a significant difference ($P < 0.05$).

reduce adiponectin production in dairy cows during early lactation.

At least in nonruminants, the differentiation of preadipocytes into mature adipocytes is processed through multistep gene regulation that is mainly driven by PPARγ and C/EBPα (Linhart et al., 2001; Kamble et al., 2020). In the present study, evident lipid droplets, as well as sustained expression of *PPARG* and *CEBPA*, indicated that bovine preadipocytes differentiated into mature adipocytes. Synthesis and secretion of adiponectin occurred specifically in mature adipocytes (Körner et al., 2005; Martella et al., 2014), which was

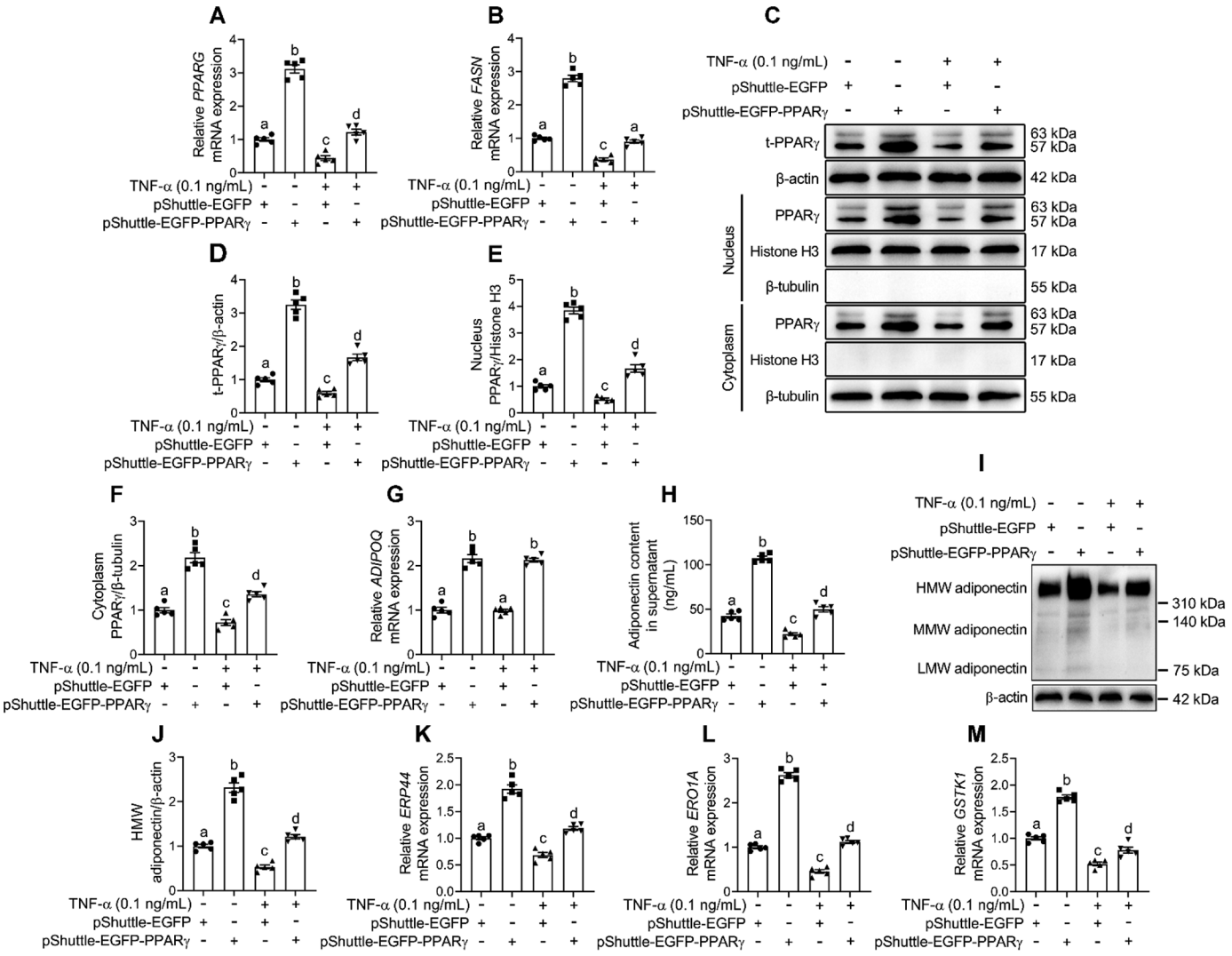



**Figure 4.** Effects of overexpression of peroxisome proliferator-activated receptor-γ (*PPARG*) on adiponectin production in tumor necrosis factor-α (TNF-α)-treated calf adipocytes. Adipocytes were transfected with pShuttle-EGFP-PPARγ or pShuttle-EGFP for 48 h after 12 d of differentiation, and then treated with or without 0.1 ng/mL TNF-α for 24 h. (A, B) Relative mRNA abundance of *PPARG* and fatty acid synthase (*FASN*) in calf adipocytes and expressed as fold change relative to control (transfection with pShuttle-EGFP) group. (C) Representative blots of total (t), nuclear, and cytoplasmic PPARγ. (D–F) Quantification of protein levels of t-PPARγ, nuclear PPARγ, and cytoplasmic PPARγ. (G) Relative mRNA abundance of adiponectin (*ADIPOQ*) in calf adipocytes and expressed as fold change relative to the control group (transfection with pShuttle-EGFP). (H) The content of adiponectin in the supernatant of adipocytes. (I) Molecular weight distribution of adiponectin in the supernatant of adipocytes. HMW = high molecular weight; MMW = middle molecular weight; LMW = low molecular weight. (J) Relative protein levels of HMW adiponectin in the supernatant of adipocytes. (K–M) Relative mRNA abundance of endoplasmic reticulum protein 44 (*ERP44*), ER oxidoreductase 1α (*ERO1A*), and disulfide bond-forming oxidoreductase A-like protein (*GSTK1*) in calf adipocytes and expressed as fold change relative to control (transfection with pShuttle-EGFP) group. Data are expressed as the means ± SEM. Data were analyzed by 2-way ANOVA with subsequent Bonferroni correction. The same letter (a–d) indicates no significant difference ($P > 0.05$), whereas different letters indicate a significant difference ($P < 0.05$).

also observed in differentiated calf adipocytes. Notably, calf adipocytes could produce adiponectin in an intralipid-free culture system. To the contrary, in the study of Krumm et al. (2018), differentiated adipocytes of adult cows secreted adiponectin into the medium only in the presence of intralipid. Because age and metabolic status affect the physiological function of adipocytes (Kahn et al., 2019; Gao et al., 2020), the discrepancy of these observations could possibly be due to the differences in sources of preadipocytes.

Studies have shown that the content of adiponectin in the supernatant of 3T3-L1 adipocytes was reduced after TNF-α treatment (Hajri et al., 2011; He et al., 2016). Consistent with these studies, we found that TNF-α lessened total and HMW adiponectin content in the supernatant of calf adipocytes. During the transi-

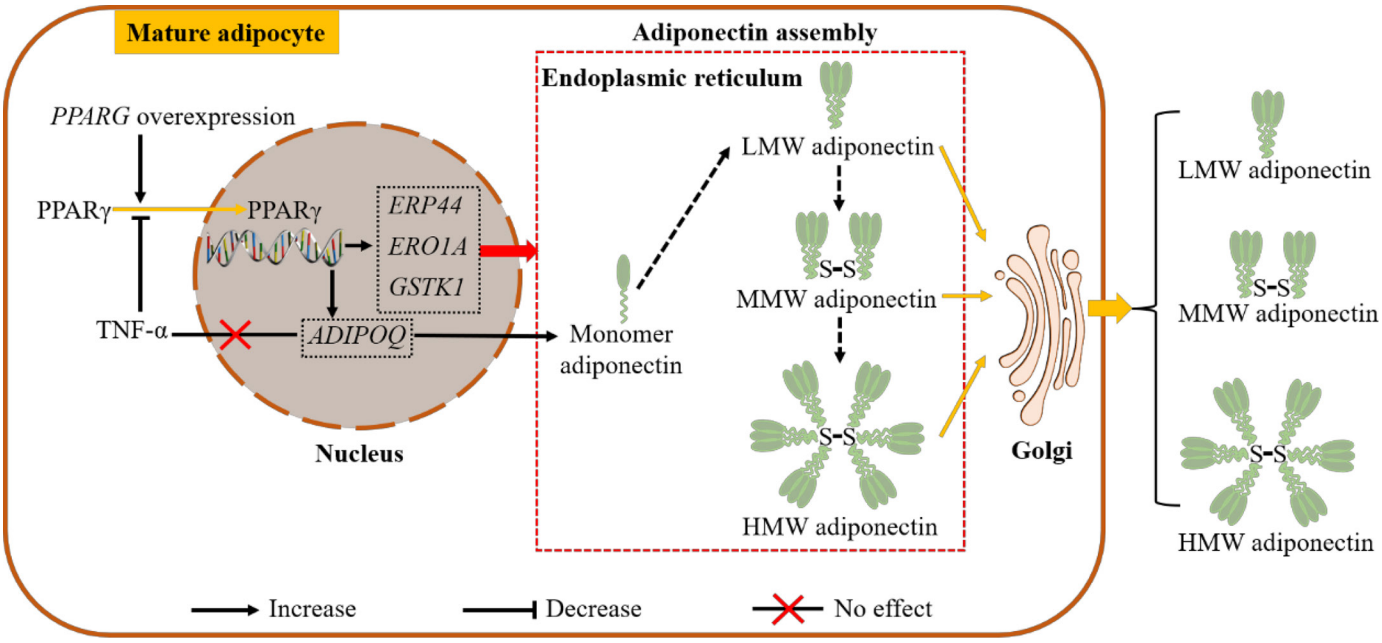



**Figure 5.** Tumor necrosis factor-α (TNF-α) reduces adiponectin production by decreasing transcriptional activity of peroxisome proliferator-activated receptor-γ (PPARγ) in calf adipocytes. TNF-α did not alter the mRNA abundance of adiponectin (*ADIPOQ*). However, TNF-α decreased PPARγ nuclear localization, which in turn downregulated the mRNA abundance of endoplasmic reticulum (ER) resident chaperones ER protein 44 (*ERP44*), ER oxidoreductase 1α (*ERO1A*), and disulfide bond-forming oxidoreductase A-like protein (*GSTK1*), resulting in reducing assembly of adiponectin monomer to low molecular weight (LMW), middle molecular weight (MMW), and high molecular weight (HMW) adiponectin. The 3 adiponectin multimers were then transferred to the Golgi for further modification and eventual extracellular release. Overexpression of *PPARG* increased nuclear localization of PPARγ, upregulated the expression of *ADIPOQ* and ER resident chaperones, and further enhanced adiponectin production.

tion period in dairy cows, elevated serum TNF-α levels often manifest together with reduced serum adiponectin levels (Mecitoglu et al., 2016). In addition, the infiltration of macrophages, a main producer of TNF-α, in AT was increased in early lactation (Newman et al., 2019). Thus, the locally elevated TNF-α concentration in AT may be one of the causal factors for the decreased adiponectin secretion in perinatal dairy cows.

A previous study demonstrated that the PPARγ agonist rosiglitazone increased serum adiponectin levels in obese mice (Tsuchida et al., 2005). In the current study, overexpression of *PPARG* elevated adiponectin production, whereas knockdown of *PPARG* decreased it, hinting that PPARγ may be responsible for adiponectin synthesis and secretion in calf adipocytes. In addition, PPARγ was found to induce the expression of C/EBPα, which in turn bound to the promoter region of *PPARG*, resulting in a self-regulatory loop (Tontonoz and Spiegelman, 2008). Thus, reduced nuclear translocation of PPARγ and downregulated mRNA abundance of *PPARG* and its downstream target gene *FASN* in TNF-α-treated calf adipocytes verified that TNF-α inhibited PPARγ transcriptional activity. Moreover, overexpression of *PPARG* alleviated TNF-α-induced reduction of adiponectin assembly and secretion in calf adipocytes, which is consistent with results from mouse adipocytes (Maeda et al., 2001; Lim et al., 2008). The present results combined with previous studies indicate that TNF-α may decrease adiponectin production in calf adipocytes partly by inhibiting the transcriptional activity of PPARγ.

Although it is clear that activation of PPARγ promotes the expression and secretion of adiponectin in human and mouse adipocytes (Iwaki et al., 2003; Seo et al., 2004; Hammarstedt et al., 2005), 0.1 ng/mL TNF-α did not change *ADIPOQ* gene expression, whereas it reduced PPARγ transcriptional activity, in calf adipocytes. In differentiated mouse adipocytes, forkhead box O1 (**FoxO1**) was reported to upregulate *ADIPOQ* gene expression by forming a transcriptional complex with C/EBPα and binding to the *ADIPOQ* promoter (Qiao and Shao, 2006). Notably, TNF-α was previously reported to enhance transcriptional activity of FoxO1 in 3T3-L1 adipocytes (Zheng et al., 2011). Thus, FoxO1 might compensate for the decreased transcriptional activity of PPARγ in TNF-α-treated calf adipocytes. However, this possibility was not analyzed in the present study and needs further elucidation.

In the ER, assembly of adiponectin monomers into multimers relies on GRP78-assisted hydrophobic inter-

actions (Wang and Scherer, 2008), ERP44-modulated thiol-mediated retention (Wang et al., 2007), and DsbA-L and Ero1-Lα-triggered disulfide bond formation (Liu et al., 2008; Fang and Judd, 2018). After assembly, multimeric forms of adiponectin enter the Golgi for further modification, with extracellular secretion ultimately depending on GGA1-coated vesicles (Xie et al., 2006). Given that TNF-α reduced HMW adiponectin production without altering mRNA abundance of *ADIPOQ*, we hypothesized that TNF-α may modulate adiponectin assembly in calf adipocytes. Indeed, previous studies in nonruminants have consistently demonstrated the suppression effects of TNF-α on Ero1-Lα and DsbA-L expression (He et al., 2016). In this study, we reported that TNF-α reduced gene expression of *ERP44*, *ERO1A*, and *GSTK1*, suggesting that TNF-α modulates adiponectin production through a posttranscriptional regulation mechanism. Inconsistent with the results of Krumm et al. (2018), which were based on using 10 ng/mL TNF-α to stimulate adult calf adipocytes, our data revealed that TNF-α (0.1 ng/mL) had no effect on *HSPA5* gene expression in calf adipocytes. This discrepancy might result from the differences in TNF-α concentration and adipocyte sources between the 2 studies. Additionally, an unexpected finding that TNF-α did not affect *GGA1* gene expression implied that TNF-α may not be involved in Golgi-mediated adiponectin export from calf adipocytes. This finding is inconsistent with the study of Lightfoot et al. (2015), who reported that TNF-α promotes Golgi-mediated release of cytokines by C2C12 mouse myoblasts. These contradictory results may be due to differences in species or cell types, and further studies are needed to explore the mechanisms underlying this finding. Together, these results suggest that TNF-α-reduced adiponectin production could be partly related to disturbance of adiponectin assembly in ER.

The finding that activation of PPARγ elevates serum HMW adiponectin levels in rodents and humans revealed that PPARγ is involved in adiponectin assembly (Hammarstedt et al., 2005; Tsuchida et al., 2005). At least in nonruminants, PPARγ agonist increased the expression of adiponectin assembly regulators, such as ERP44, Ero1-Lα, and DsbA-L (Wang et al., 2007; Jin et al., 2015). Consistently, our results revealed that overexpression of *PPARG* upregulated *ERP44*, *ERO1A*, and *GSTK1* expression and partially reversed the inhibition effects of TNF-α on HWM adiponectin secretion in calf adipocytes. Thus, our findings together with previous studies provide mechanistic insight that TNF-α controls adiponectin assembly in calf adipocytes, at least in part, by modulating PPARγ transcriptional activity.

Our results revealed that overexpression of *PPARG* attenuated the inhibitory effects of TNF-α on adiponectin production in calf adipocytes. Kushibiki et al. (2001) found that TNF-α-induced insulin resistance was partially reversed by 2,4-thiazolidinedione, a potent PPARγ agonist (Sung et al., 2006), in dairy steers. Similarly, rosiglitazone, another PPARγ agonist, was reported to increase systemic insulin sensitivity and reduce hepatic steatosis in patients with nonalcoholic steatohepatitis (Neuschwander-Tetri et al., 2003). Notably, postpartum dairy cows often experience insulin resistance and hepatic lipid accumulation, which may lead to the onset of ketosis and fatty liver (Andrews, 1998; De Koster and Opsomer, 2013). Therefore, activation of PPARγ may be an effective therapeutic approach to improve the metabolic disorder of periparturient dairy cows.

In the present study, preadipocytes were isolated from AT of the calves. Tamate et al. (1962) found that all 4 compartments of the ruminant stomach except abomasum are nonfunctional and small in size at birth, suggesting that calves are functionally monogastric. Therefore, differences may exist in the metabolic status between a calf and a perinatal dairy cow. Additionally, the differentiation capacity of adipocytes from elderly and young rats was previously shown to be nonidentical, and the adipogenicity of adipocytes from calf and adult dairy cows might exhibit a similar discrepancy (Schipper et al., 2008; Aird et al., 2015). Further mechanistic research on the role of TNF-α in regulating adiponectin secretion in adult bovine adipocytes are needed.

## CONCLUSIONS

TNF-α decreased PPARγ transcriptional activity and reduced adiponectin production in calf adipocytes. Overexpression of *PPARG* partially reversed the inhibition effects of TNF-α on adiponectin production, whereas knockdown of *PPARG* reduced adiponectin production in calf adipocytes. Thus, TNF-α-inhibited transcriptional activity of PPARγ might be partly responsible for decreased adiponectin levels in dairy cows during the peripartal period.

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